

Elasticity

Based on Ch. 4, Economics 10th ed., Parkin, Powell & Mathews

Course: Eco 101 (Introduction to Microeconomics)

Price Elasticity of Demand (PED)

The law of demand states that a fall in the price of a good raises the quantity demanded. The **price elasticity of demand** measures how much the quantity demanded **responds** to a change in the price.

$$E_d = \frac{\text{Percentage change in quantity demand}}{\text{Percentage change in price}} = \frac{\% \Delta Q_d}{\% \Delta P}$$

In the formula, E_d = coefficient of price elasticity of demand, or simply elasticity coefficient; % =percentage; and Δ stands for “change in.”

For example, suppose that after a 10 percent increase in the price of an ice-cream cone, you buy 20 percent fewer cones. We calculate your elasticity of demand this way:

$$\text{Price elasticity of demand} = \frac{20 \text{ percent}}{10 \text{ percent}} = 2.$$

The Midpoint Method: A Better Way to Calculate Percentage Changes and Elasticities

The midpoint formula for elasticity calculates the price elasticity of demand using the average of initial and final values of price and quantity

$$\text{Price elasticity of demand} = \frac{(Q_2 - Q_1) / [(Q_2 + Q_1) / 2]}{(P_2 - P_1) / [(P_2 + P_1) / 2]}$$

- Example: If the price of an ice cream cone increases from \$2.00 to \$2.20 and the amount you buy falls from 10 to 8 cones, then your elasticity of demand, using the midpoint formula, would be calculated as:

$$\frac{\frac{(10 - 8)}{(10 + 8) / 2}}{\frac{(2.20 - 2.00)}{(2.00 + 2.20) / 2}} = \frac{22\%}{9.5\%} = 2.32$$

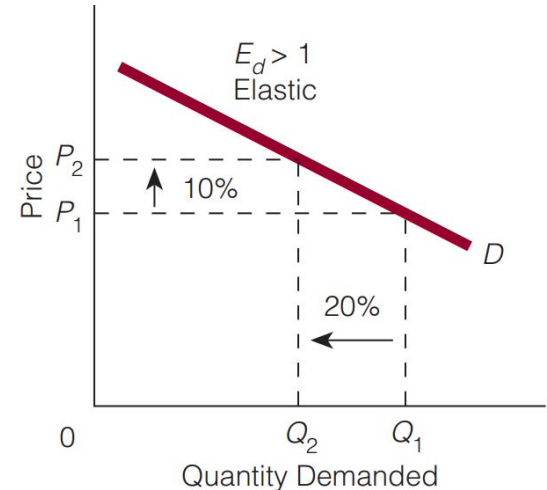
Variety of Demand Elasticity

1. ELASTIC DEMAND ($E_d > 1$):

Percentage change in quantity demanded $>$ Percentage change in price $\rightarrow$
 $E_d > 1 \rightarrow$ Demand is elastic

When $E_d > 1$, demand is **elastic**, meaning the quantity demanded changes more than the price does.

For example, a 10% price increase leads to a 20% drop in quantity demanded ($E_d = 2$).



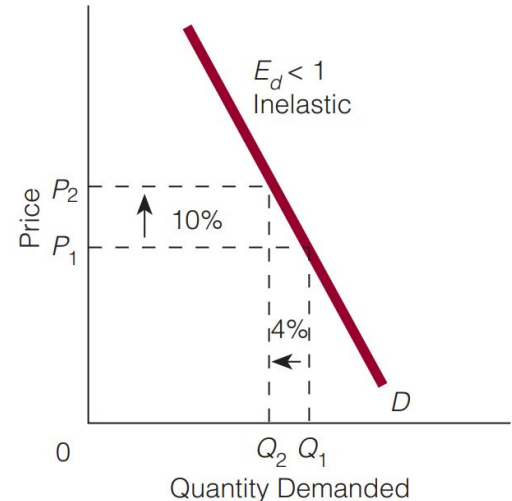
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2. INELASTIC DEMAND ($E_d < 1$):

Percentage change in quantity demanded $<$ Percentage change in price $\rightarrow$
 $E_d < 1 \rightarrow$ Demand is inelastic

When $E_d < 1$, demand is **inelastic**, meaning the quantity demanded changes less than the price does.

For example, a 10% price increase leads to a 4% drop in quantity demanded ($E_d = 0.4$).



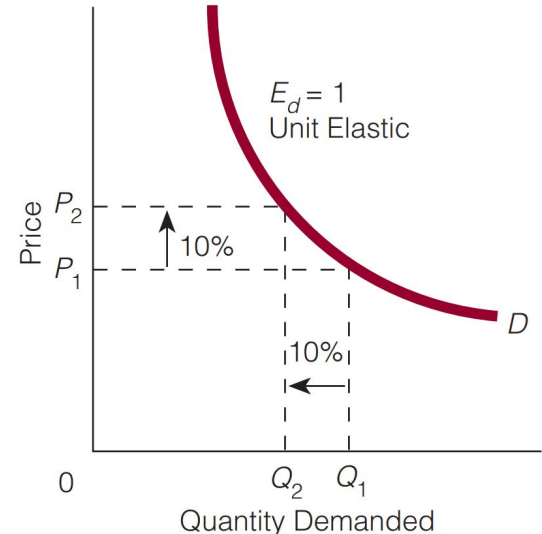
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3. UNIT ELASTIC DEMAND ($E_d = 1$):

Percentage change in quantity demanded = Percentage change in price $\rightarrow$
 $E_d = 1 \rightarrow$ Demand is unit elastic

When $E_d = 1$, demand is **unit elastic**, meaning the quantity demanded changes proportionately with the price.

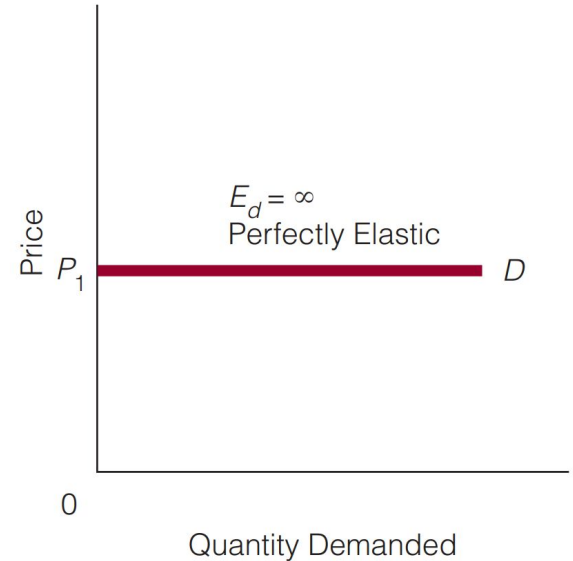
For example, a 10% price increase leads to a 10% drop in quantity demanded ($E_d = 1$).



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4. PERFECTLY ELASTIC DEMAND ($E_d = \infty$)

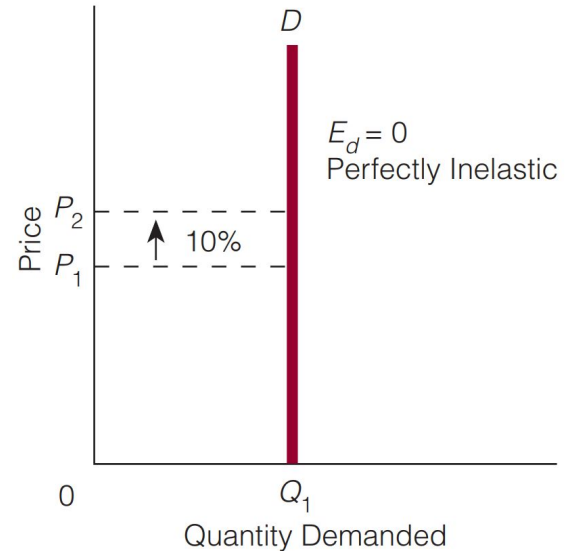
- Occurs where even a tiny price change leads to an **extreme change** in quantity demanded, from buying everything to buying nothing.
- For example, suppose buyers are willing to buy all units of a seller's good at \$5 per unit but nothing at \$5.10



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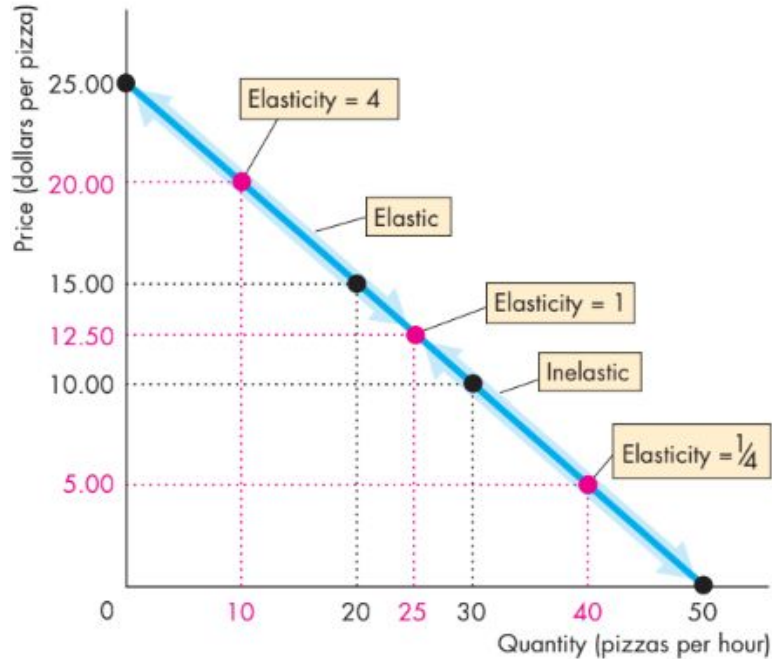
5. PERFECTLY INELASTIC DEMAND ($E_d=0$)

- When quantity demanded **does not change** as price changes
- For example, if the price of Dogs Love It dog food rises by 10% (from \$2 to \$2.20) and Jeremy continues buying the same amount, the price change has no effect on the quantity demanded. Jeremy's demand is perfectly inelastic between \$2 and \$2.20.



Elasticity Along a Linear Demand Curve

FIGURE 4.4 Elasticity Along a Linear Demand Curve



On a linear demand curve, demand is unit elastic at the midpoint (elasticity is 1), elastic above the midpoint, and inelastic below the midpoint.

Total Revenue & Elasticity

The **total revenue (TR)** of a seller equals the price of a good times the quantity of the good sold.

$$\text{Total Revenue} = \text{Price} \times \text{Quantity}$$

When price changes, total revenue also changes. But a cut in the price does not always decrease total revenue. The change in total revenue depends on the elasticity of demand in the following way:

- If **demand is elastic**, a 1 percent price cut increases the quantity sold by **more than 1** percent and **total revenue increases**.
- If **demand is inelastic**, a 1 percent price cut increases the quantity sold by **less than 1** percent and **total revenue decreases**.
- If **demand is unit elastic**, a 1 percent price cut increases the quantity sold by 1 percent and **total revenue does not change**.

Numerical examples to help understand the relationship between **price elasticity of demand** and **total revenue** ($TR = \text{Price} \times \text{Quantity}$).

1. Elastic Demand

- **Original Price** = \$10
- **Original Quantity** = 100 units
- **Total Revenue** = $\$10 \times 100 = \$1,000$

Now **price drops by 10%** → New Price = \$9

Because demand is **elastic**, **quantity increases by 20%** → New Quantity = 120 units

- **New Total Revenue** = $\$9 \times 120 = \$1,080$ (**Total revenue increases**)

2. Inelastic Demand

- **Original Price** = \$10
- **Original Quantity** = 100 units
- **Total Revenue** = \$1,000

Price drops by 10% → New Price = \$9

Because demand is **inelastic**, quantity increases by only 5% → New Quantity = 105 units

- **New Total Revenue** = $\$9 \times 105 = \945 (**Total revenue decreases**)

3. Unit Elastic Demand

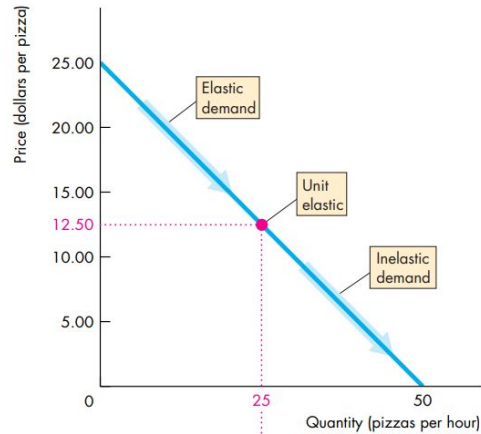
- **Original Price** = \$20
- **Original Quantity** = 50 units
- **Total Revenue** = $\$20 \times 50 = \$1,000$

Now halve the price → New Price = **\$10**

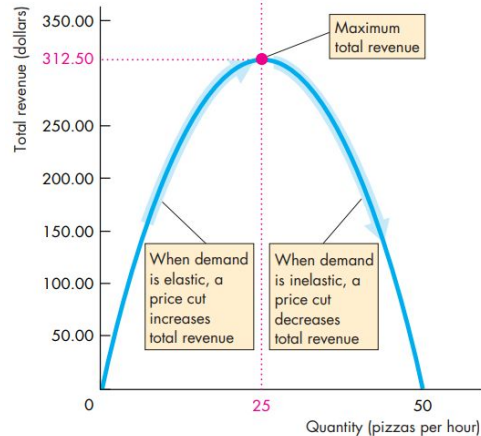
To keep total revenue constant, quantity must **double** (because $50 \times 2 = 100$)

- New Quantity = **100 units**
- New Total Revenue = $\$10 \times 100 =$
\$1,000 (Total Revenue Unchanged)

FIGURE 4.5 Elasticity and Total Revenue



(a) Demand



(b) Total revenue

When demand is elastic, in the price range from \$25 to \$12.50, a decrease in price (part a) brings an increase in total revenue (part b). When demand is inelastic, in the price range from \$12.50 to zero, a decrease in price (part a) brings a decrease in total revenue (part b). When demand is unit elastic, at a price of \$12.50 (part a), total revenue is at a maximum (part b).

Factors Affecting Price Elasticity of Demand

1. The Closeness of Substitutes

If a good has **many close substitutes**, demand is **elastic**.
→ People can easily switch to something else if the price goes up.

If a good has **few or no substitutes**, demand is **inelastic**.
→ People have no choice but to keep buying it, even if the price rises.

Necessities vs. Luxuries

- **Necessity** = Something you *must* have (e.g., medicine, salt)
→ Has **few substitutes** → **Inelastic demand**
- **Luxury** = Something you *can live without* (e.g., vacation, designer bag)
→ Has **many substitutes** (or you can just skip it) → **Elastic demand**

Factors Affecting Price Elasticity of Demand (Cont.)

2. Percentage of One's Budget Spent on the Good

If a good takes up a **large portion** of your budget → **Elastic demand**
→ Even a small price change matters a lot, so people may buy less.

If a good takes up a **small portion** of your budget → **Inelastic demand**
→ Price changes don't affect much, so people keep buying it.

Example to Help Remember:

- **New laptop** = Expensive → Big part of your budget → **Elastic demand**
- **Salt or matchbox** = Cheap → Tiny part of your budget → **Inelastic demand**

Factors Affecting Price Elasticity of Demand (Cont.)

3. Time

- In the **short run**, demand is usually **inelastic**.
→ People need time to adjust their habits or find alternatives.
- In the **long run**, demand becomes more **elastic**.
→ With time, people can change their behavior, switch products, or plan better.

Example:

- If fuel prices go up today, you still need to drive → **Short run = Inelastic**
- Over time, you might use public transport or buy a fuel-efficient car → **Long run = Elastic**

Cross Elasticity of Demand (XED)

Cross elasticity of demand measures the responsiveness in the quantity demanded of one good to changes in the price of another good.

$$E_c = \text{Cross elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price of a substitute or complement}}$$

where E_c stands for the coefficient of cross elasticity of demand, or the elasticity coefficient

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This concept is often used to determine whether goods are **substitutes** or **complements** and the extent of their relationship

For example, the price of pizza is constant and people buy 9 pizzas an hour. Then the price of a burger rises from \$1.50 to \$2.50. No other influence on buying plans changes and the quantity of pizzas bought increases to 11 an hour.

Then, the cross elasticity of demand for pizza with respect to the price of burger is:

$$E_c = \frac{\Delta Q/Q_{ave} \times 100 = (+2/10) \times 100 = +20\%}{\Delta P/P_{ave} \times 100 = (+ \$1/\$2) \times 100 = +50\%}.$$
$$E_c = \frac{+20\%}{+50\%} = 0.4.$$

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When the cross elasticity of demand is **positive**, the quantity demanded of one good increases as the price of another good rises. This indicates that the goods are **substitutes**.

$E_c > 0 \rightarrow$ the two goods are substitutes

Continued

When the cross elasticity of demand is **negative**, the quantity demanded of one good decreases as the price of another good rises. This indicates that the goods are **complements**.

$E_c < 0$ → the two goods are complements

For example, if the price of pizza is constant and 11 pizzas an hour are bought. Then the price of a soft drink rises from \$1.50 to \$2.50. No other influence on buying plans changes and the quantity of pizzas bought falls to 9 an hour. The quantity of pizzas demanded decreases by 20 percent (−20%). The price rises by 50 percent (+50%). So the cross elasticity of demand for pizza with respect to the price of a soft drink is:

$$E_c = \frac{-20\%}{+50\%} = -0.4.$$

This shows that pizzas and soft drinks are complements.

Income Elasticity Of Demand (YED)

Income elasticity of demand measures the responsiveness of quantity demanded to changes in income.

$$E_y = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

where E_y denotes the coefficient of income elasticity of demand, or the elasticity coefficient.



Continued

Income elasticities of demand can be positive or negative and they fall into three interesting ranges:

■ Greater than 1 (normal good, income elastic)

- *Income Elastic Demand*: The quantity demanded increases by a higher percentage than the percentage increase in income, indicating that the demand for the good is income elastic.

■ Positive and less than 1 (normal good, income inelastic)

- *Income Inelastic Demand*: When the income elasticity is positive but less than 1, the percentage increase in quantity demanded is less than the percentage increase in income.

$$E_y > 0 \rightarrow \text{Normal good}$$

■ Negative (inferior good)

- *Inferior Goods*: If income elasticity is negative, the demand for the good decreases as income rises, such as with small motorcycles or potatoes.

$$E_y < 0 \rightarrow \text{Inferior good}$$

Example

A person's income increases from \$500 to \$600 per month and, as a result, the quantity demanded of good X increases from 20 units to 30 units per month. Then we have income elasticity of 2.2, as shown here:

$$E_y = \frac{\frac{10}{25}}{\frac{100}{550}} = 2.2$$

Since E_y is a **positive** number, so good X is a **normal** good.

Price Elasticity Of Supply (PES)

- Price elasticity of supply measures the responsiveness of quantity supplied to changes in price

$$E_s = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

In the formula, E_s = coefficient of price elasticity of supply, or simply elasticity coefficient; % =percentage; and Δ stands for “change in.”

- It measures how willing consumers are to buy less of the good as its price rises or it measures the price sensitivity of buyer's demand

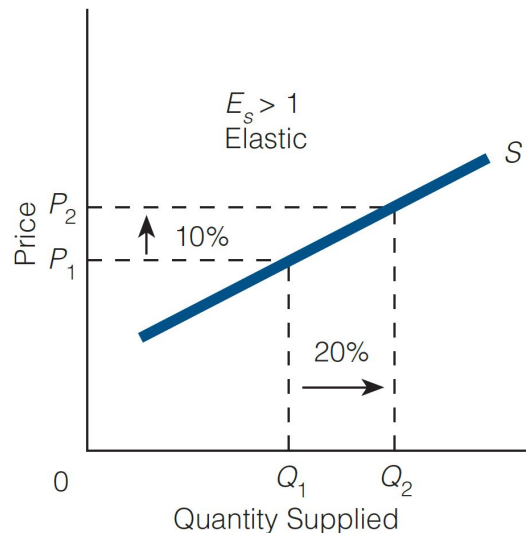
Variety of Supply Elasticity

1. ELASTIC SUPPLY ($E_s > 1$):

Percentage change in quantity supplied $>$ Percentage change in price $\rightarrow$
 $E_s > 1 \rightarrow$ Elastic supply

Elastic supply (**$E_s > 1$**) refers to a percentage change in quantity supplied that is greater than the percentage change in price of the good

For example, a 10% price increase leads to a 20% increase in quantity supplied ($E_s = 2$).



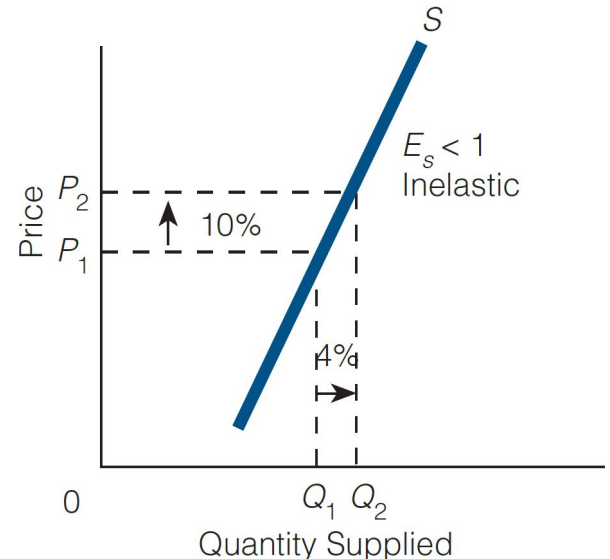
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2. INELASTIC SUPPLY ($E_s < 1$):

Percentage change in quantity supplied $<$ Percentage change in price $\rightarrow$
 $E_s < 1 \rightarrow$ Inelastic supply

Elastic supply (**$E_s < 1$**) refers to a percentage change in quantity supplied that is smaller than the percentage change in price of the good.

For example, a 10% price increase leads to a 4% increase in quantity supplied ($E_s = 0.4$).



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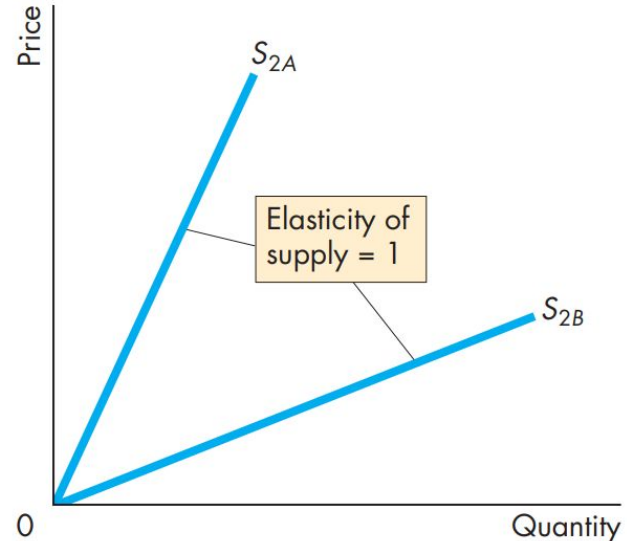
3. UNIT ELASTIC SUPPLY ($E_s = 1$):

Percentage change in quantity supplied = Percentage change in price $\rightarrow$
 $E_s = 1 \rightarrow$ Unit elastic supply

When $E_s = 1$, supply is **unit elastic**, meaning the quantity supplied changes proportionately with the price.

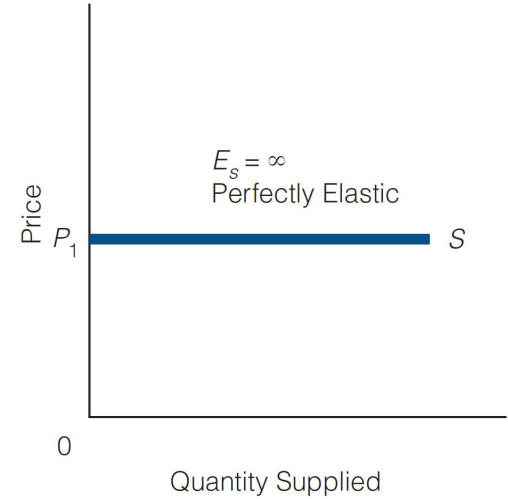
For example, a 10% price increase leads to a 10% rise in quantity supplied ($E_s = 1$).

No matter how steep the supply curve is, if it is linear and passes through the origin, supply is unit elastic.



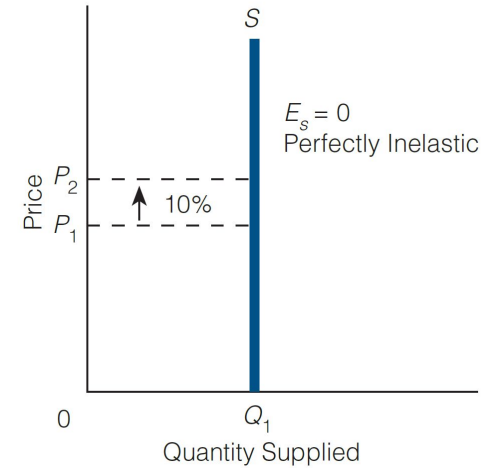
4. PERFECTLY ELASTIC SUPPLY ($E_s = \infty$)

- Occurs where even a tiny price change leads to an **infinitely large change** in quantity supplied



5. PERFECTLY INELASTIC SUPPLY ($E_s=0$)

- When quantity supplied **does not change** as price changes



The Factors That Influence the Elasticity of Supply

1. Resource Substitution Possibilities

If a good is made using **common resources**, supply is **elastic**.

→ Producers can easily shift resources to make more (e.g., wheat vs. corn).

If a good is made using **rare or unique resources**, supply is **inelastic**.

→ It's hard or impossible to produce more (e.g., Van Gogh paintings).

Example:

- **Elastic supply** → Wheat: Farmers can switch between crops easily.
- **Inelastic supply** → Original artwork: Only one artist, limited number.

2. Time Frame for the Supply Decision

1. Momentary Supply

- **Very short time (right now)**
- Supply is **perfectly inelastic** for goods like oranges — can't grow more today.
- Some services (like phone calls) can have **perfectly elastic** supply — easy to increase instantly.

2. Short-Run Supply

- **Some time has passed**
- Producers can make **limited adjustments** (e.g., work extra shifts).
- Supply is usually **inelastic**, but not perfectly.

3. Long-Run Supply

- **Plenty of time to adjust**
- Firms can change everything — build new factories, enter or exit the industry.
- Supply becomes **elastic or even perfectly elastic** for most goods.